

Scenario 1: Smart City Air Quality Monitoring

R CODE:

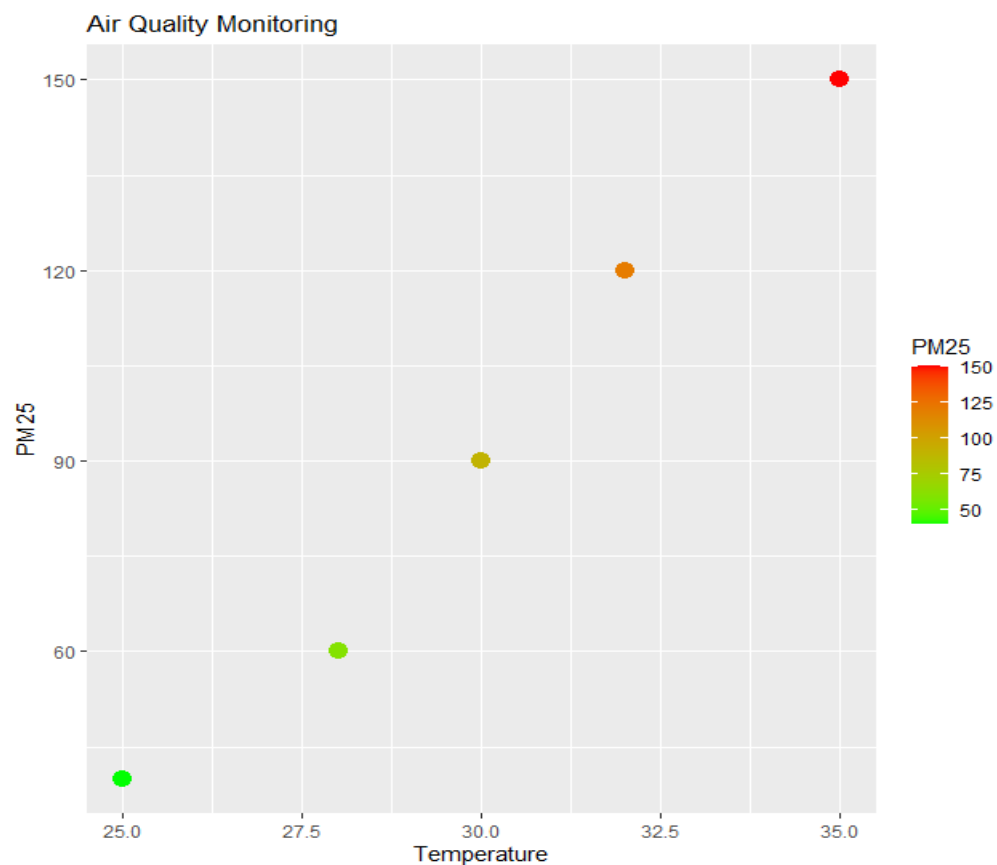
```
library(ggplot2)

Temperature <- c(25,28,30,32,35)
PM25 <- c(40,60,90,120,150)

data <- data.frame(Temperature, PM25)

ggplot(data, aes(x=Temperature, y=PM25, color=PM25)) +
  geom_point(size=4) +
  scale_color_gradient(low="green", high="red") +
  ggtitle("Air Quality Monitoring")
```

OUTPUT:



python code:

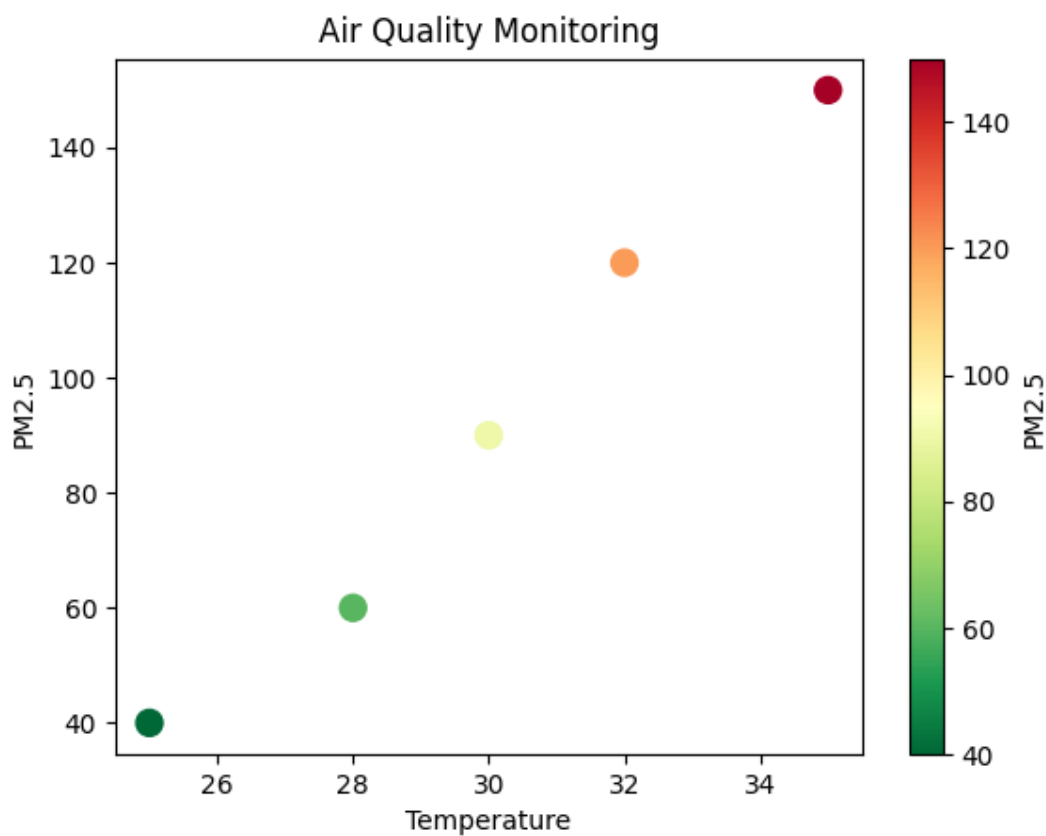
```
import matplotlib.pyplot as plt

temperature = [25,28,30,32,35]
pm25 = [40,60,90,120,150]

plt.scatter(temperature, pm25,
            c=pm25, cmap="RdYlGn_r", s=100)

plt.colorbar(label="PM2.5")
plt.title("Air Quality Monitoring")
plt.xlabel("Temperature")
plt.ylabel("PM2.5")
plt.show()
```

OUTPUT:



Scenario 2: Earthquake Risk Analysis

R CODE:

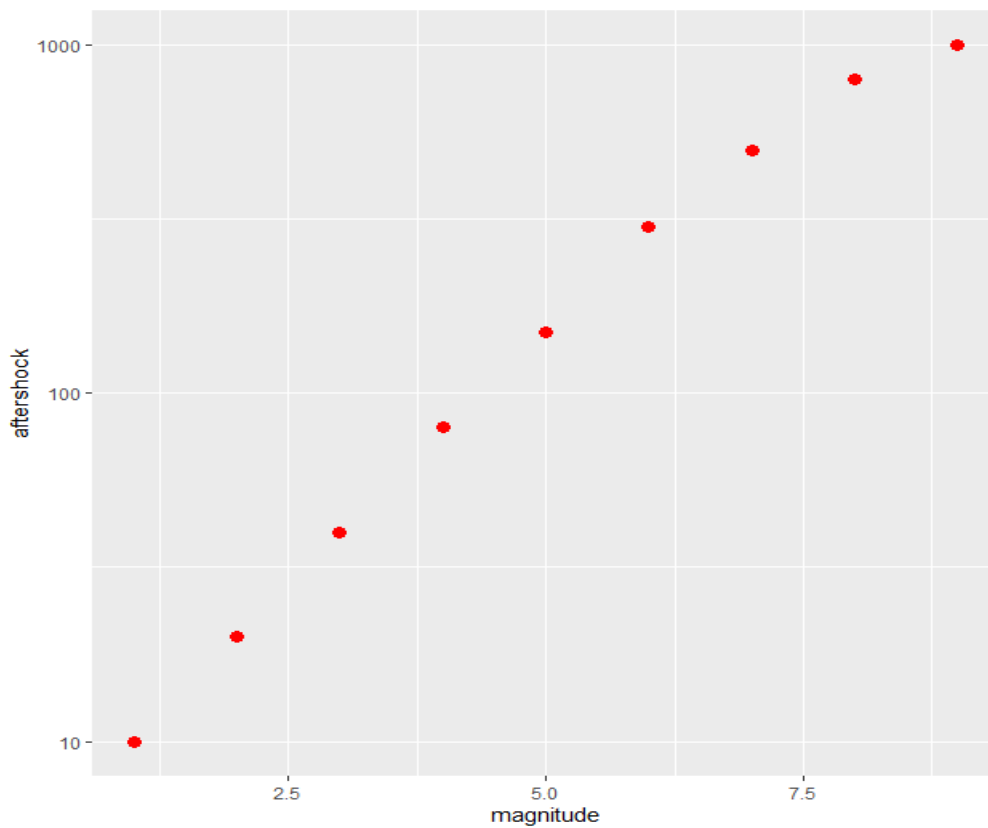
```
library(ggplot2)

magnitude <- c(1,2,3,4,5,6,7,8,9)
aftershock <- c(10,20,40,80,150,300,500,800,1000)

# Linear Scale
ggplot(data.frame(magnitude, aftershock),
       aes(magnitude, aftershock)) +
  geom_point(color="blue", size=3)

# Log Scale
ggplot(data.frame(magnitude, aftershock),
       aes(magnitude, aftershock)) +
  geom_point(color="red", size=3) +
  scale_y_log10()
```

OUTPUT:



Python code:

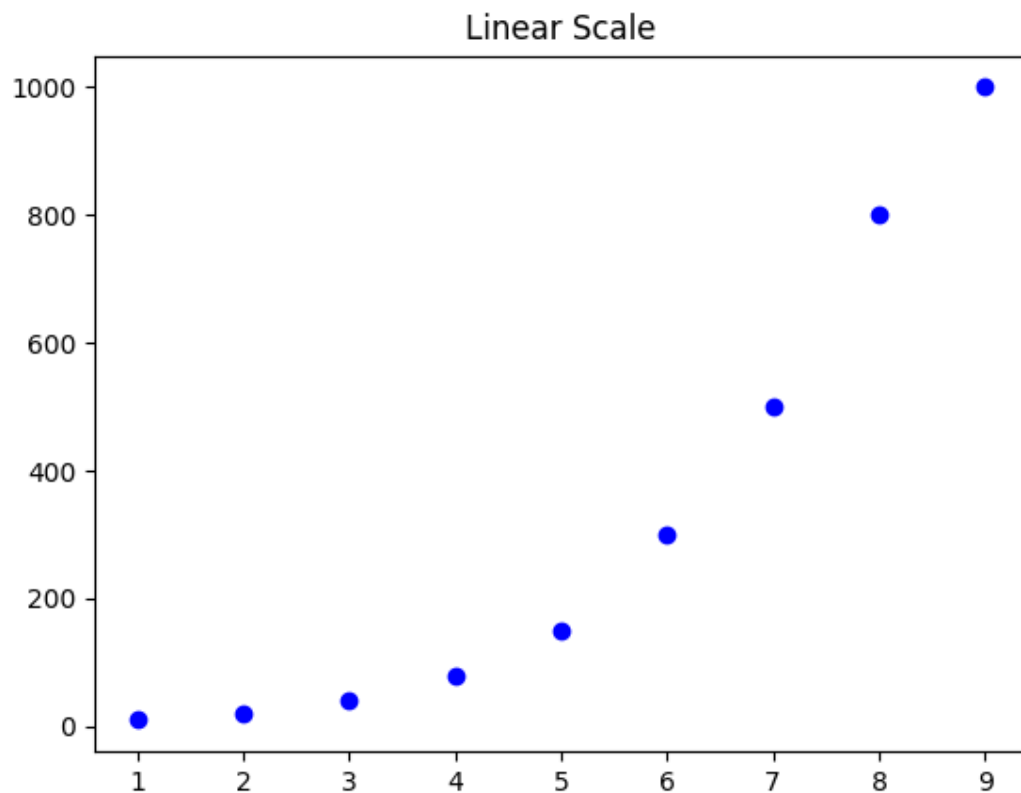
```
import matplotlib.pyplot as plt

magnitude = [1,2,3,4,5,6,7,8,9]
aftershock = [10,20,40,80,150,300,500,800,1000]

# Linear Scale
plt.scatter(magnitude, aftershock, color="blue")
plt.title("Linear Scale")
plt.show()

# Log Scale
plt.scatter(magnitude, aftershock, color="red")
plt.yscale("log")
plt.title("Log Scale")
plt.show()
```

OUTPUT:



Scenario 3: Wind Direction Analysis for Airport Operations

R code:

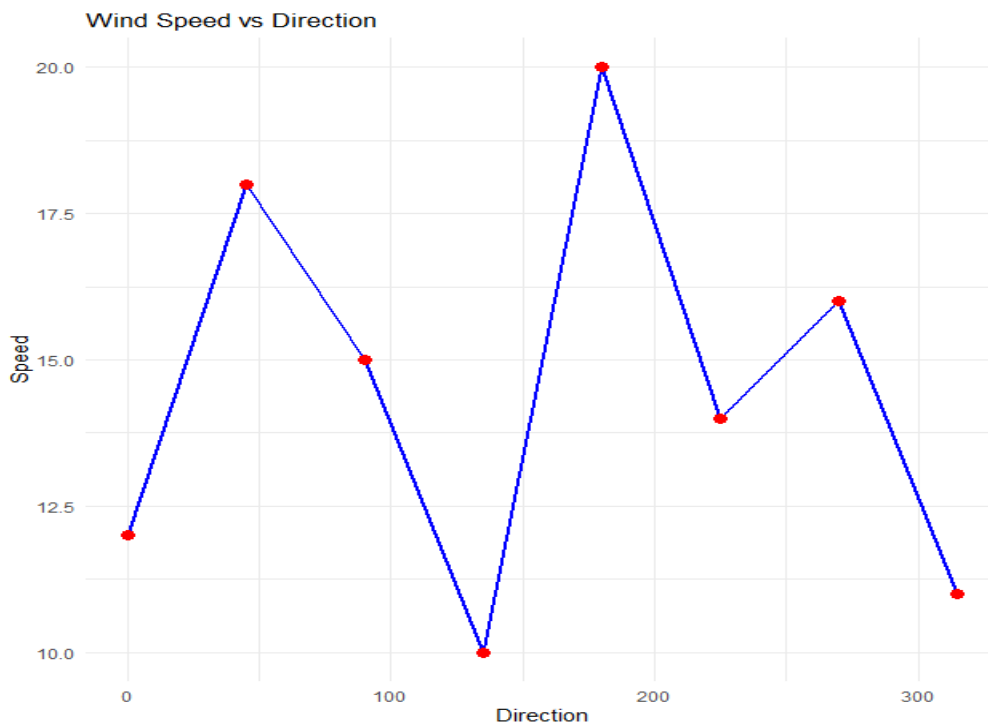
```
library(ggplot2)

wind <- data.frame(
  Direction = c(0,45,90,135,180,225,270,315),
  Speed = c(12,18,15,10,20,14,16,11)
)

# Polar (Wind Rose)
ggplot(wind, aes(x=factor(Direction), y=Speed, fill=Speed)) +
  geom_bar(stat="identity") +
  coord_polar() +
  labs(title="Wind Rose",
       x="Direction",
       y="Speed") +
  theme_minimal()

# Cartesian Plot
ggplot(wind, aes(Direction, Speed)) +
  geom_line(color="blue", linewidth=1) +
  geom_point(size=3, color="red") +
  labs(title="Wind Speed vs Direction") +
  theme_minimal()
```

OUTPUT:



Python code :

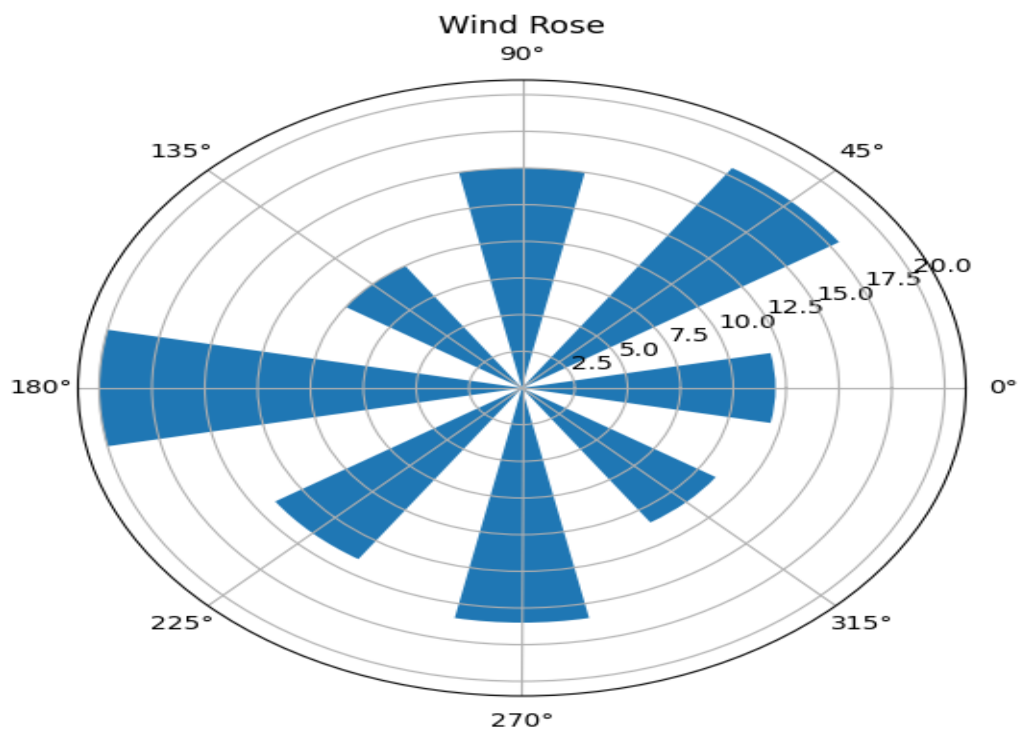
```
import matplotlib.pyplot as plt
import numpy as np

direction = np.deg2rad([0,45,90,135,180,225,270,315])
speed = [12,18,15,10,20,14,16,11]

# Polar Plot
plt.figure(figsize=(6,6))
ax = plt.subplot(111, polar=True)
ax.bar(direction, speed, width=0.4)
plt.title("Wind Rose")
plt.show()

# Cartesian Plot
plt.figure(figsize=(6,4))
plt.plot(np.rad2deg(direction), speed, marker='o')
plt.title("Wind Speed vs Direction")
plt.xlabel("Direction")
plt.ylabel("Speed")
plt.grid(True)
plt.show()
```

OUTPUT:



Scenario 4: Hospital Patient Monitoring Dashboard

R code:

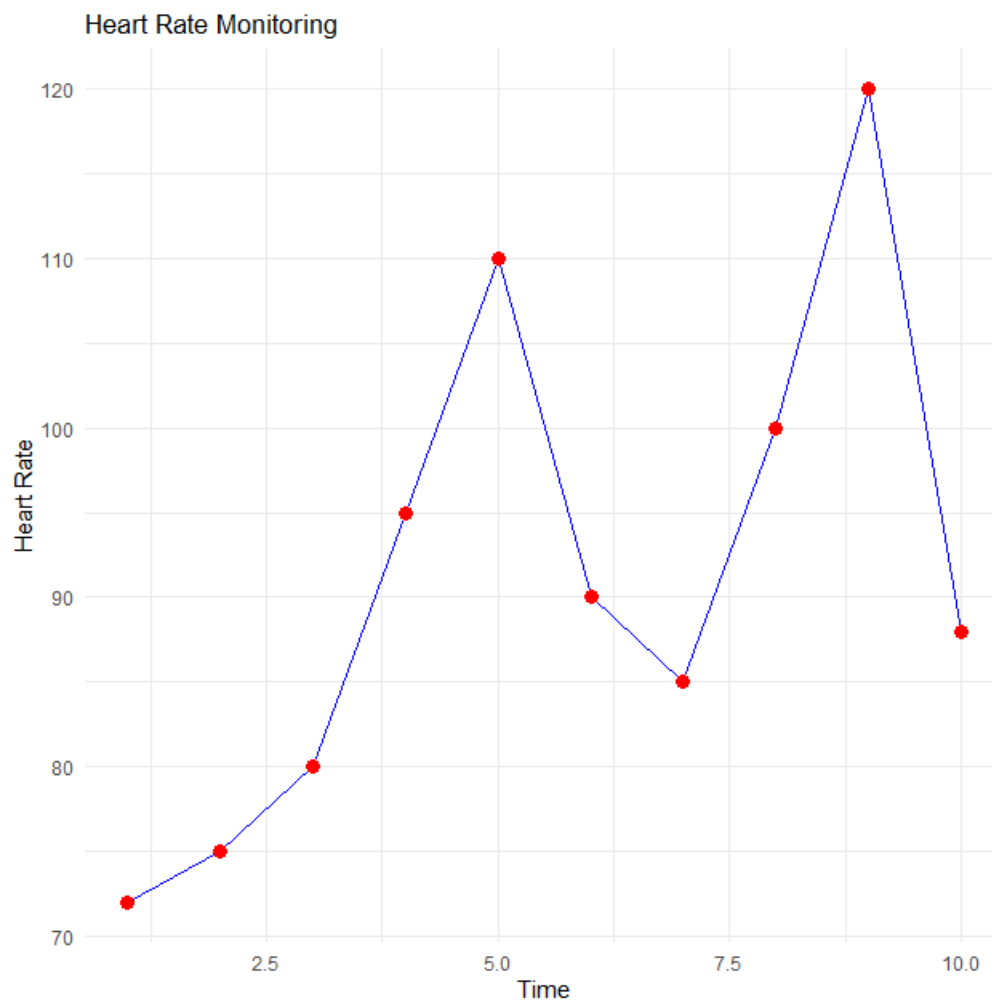
```
library(ggplot2)

time <- 1:10
heart <- c(72,75,80,95,110,90,85,100,120,88)

data <- data.frame(time, heart)

ggplot(data, aes(x=time, y=heart)) +
  geom_line(color="blue") +
  geom_point(color="red", size=3) +
  labs(title="Heart Rate Monitoring",
       x="Time",
       y="Heart Rate") +
  theme_minimal()
```

OUTPUT:



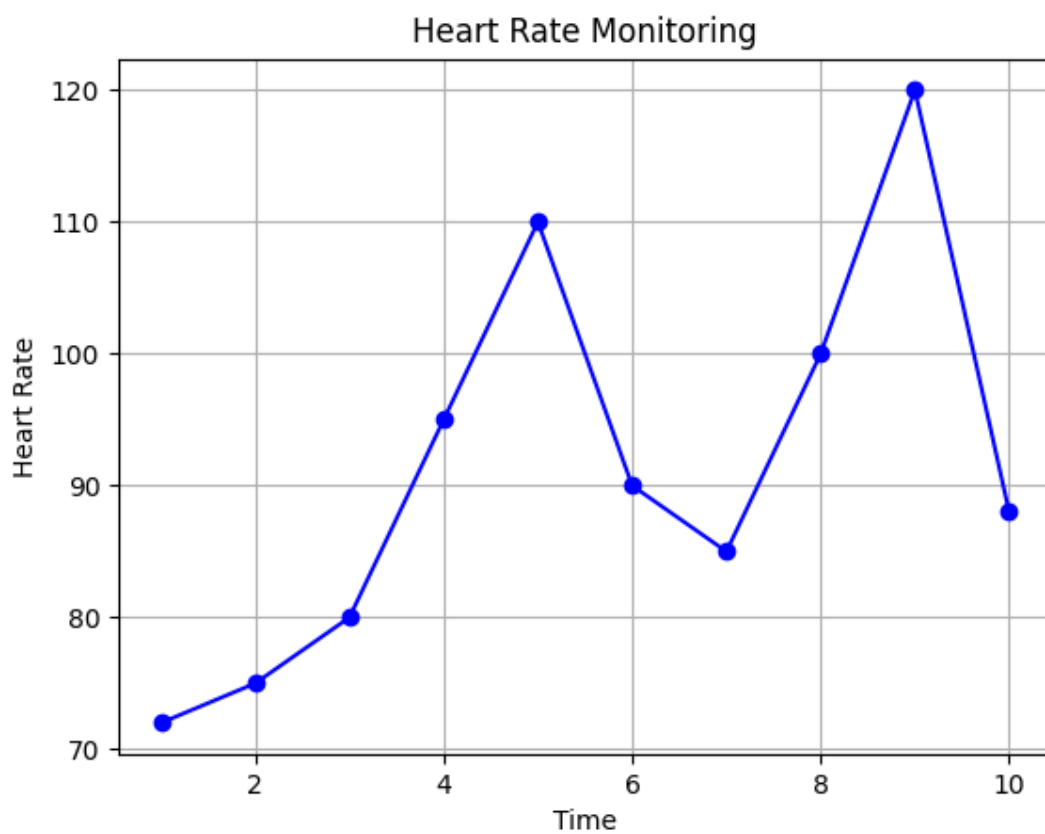
Python code :

```
import matplotlib.pyplot as plt

time = [1,2,3,4,5,6,7,8,9,10]
heart = [72,75,80,95,110,90,85,100,120,88]

plt.plot(time, heart, color="blue", marker="o")
plt.title("Heart Rate Monitoring")
plt.xlabel("Time")
plt.ylabel("Heart Rate")
plt.grid(True)
plt.show()
```

OUTPUT:



Scenario 5: Global Climate Change Study

R Code:

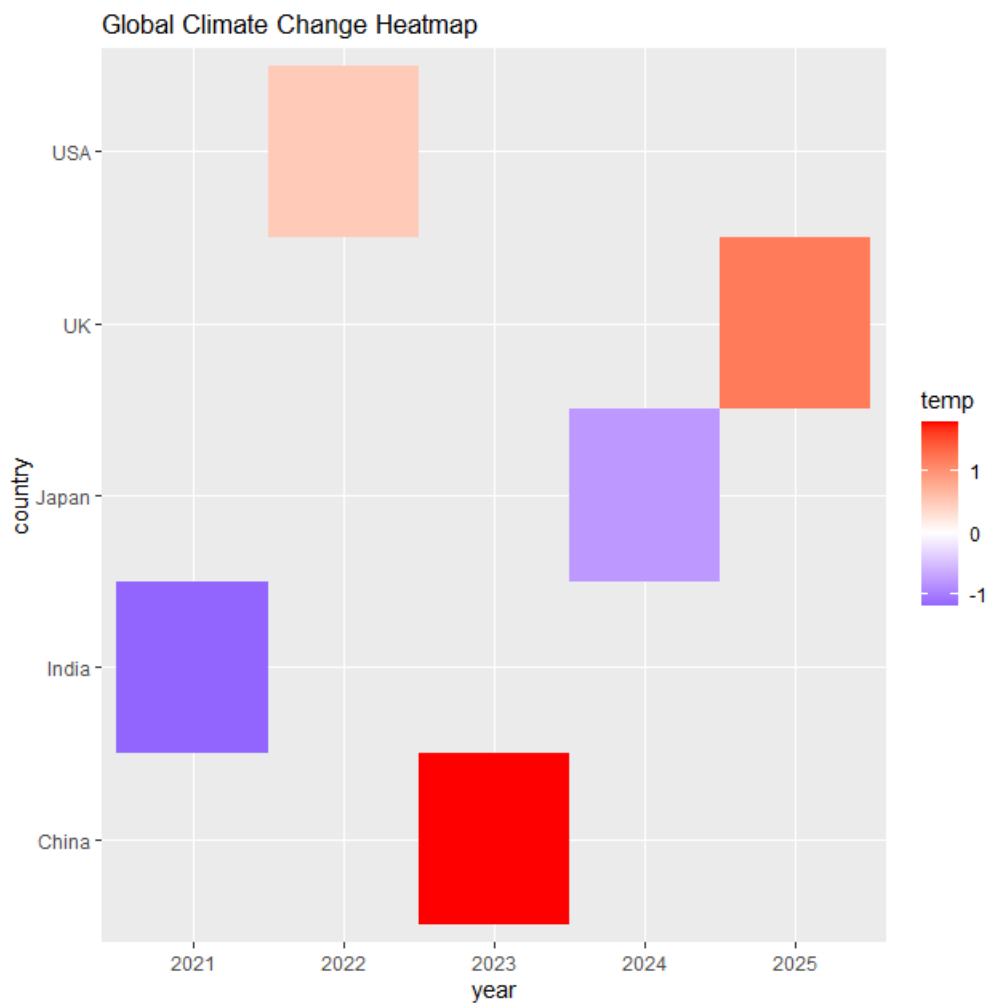
```
library(ggplot2)

country <- c("India","USA","China","Japan","UK")
year <- c("2021","2022","2023","2024","2025")
temp <- c(-1.2,0.5,1.8,-0.8,1.2)

data <- data.frame(country, year, temp)

ggplot(data, aes(x=year, y=country, fill=temp)) +
  geom_tile() +
  scale_fill_gradient2(low="blue", mid="white", high="red") +
  labs(title="Global Climate Change Heatmap")
```

OUTPUT:



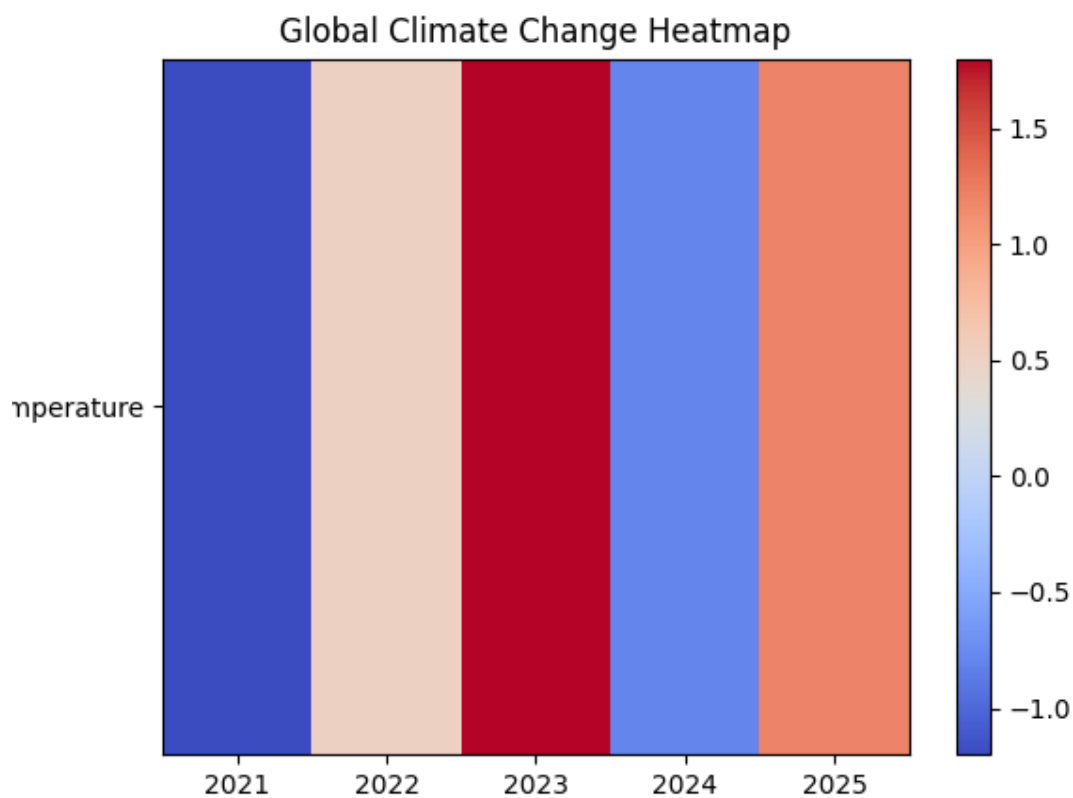
Python code:

```
import matplotlib.pyplot as plt
import numpy as np

data = np.array([[ -1.2, 0.5, 1.8, -0.8, 1.2]])

plt.imshow(data, cmap="coolwarm", aspect="auto")
plt.xticks(range(5), ["2021", "2022", "2023", "2024", "2025"])
plt.yticks([0], ["Temperature"])
plt.colorbar()
plt.title("Global Climate Change Heatmap")
plt.show()
```

OUTPUT:



Scenario 6: E-Commerce Sales Analytics

R code:

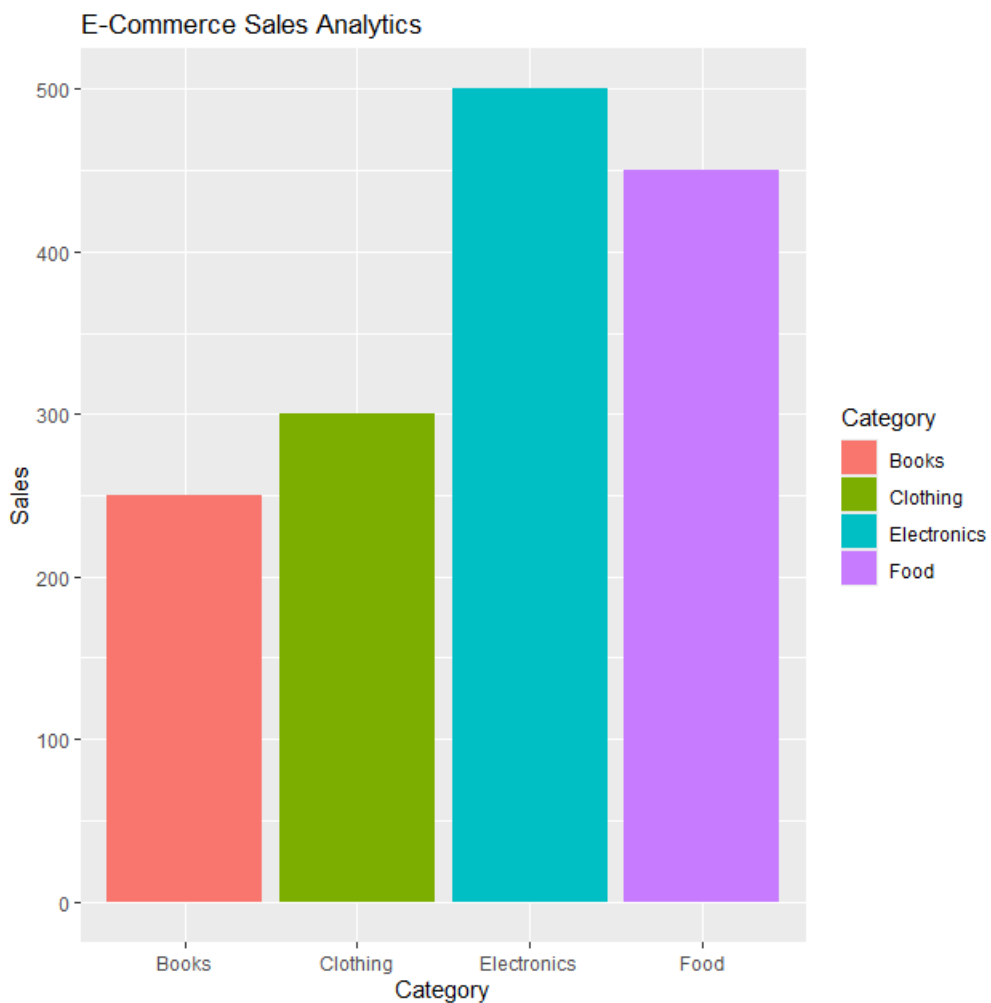
```
library(ggplot2)

Category <- c("Electronics","Clothing","Food","Books")
Sales <- c(500,300,450,250)

data <- data.frame(Category, Sales)

ggplot(data, aes(x=Category, y=Sales, fill=Category)) +
  geom_bar(stat="identity") +
  ggtitle("E-Commerce Sales Analytics") +
  xlab("Category") +
  ylab("Sales")
```

OUTPUT:



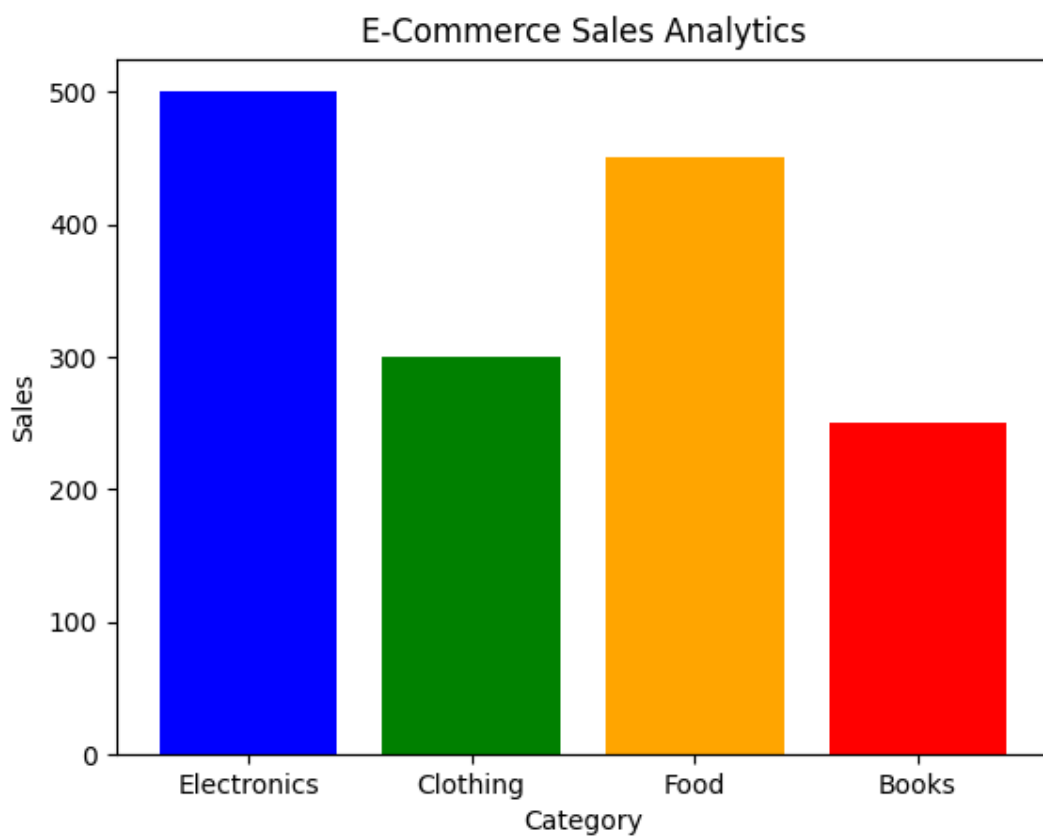
Python code:

```
import matplotlib.pyplot as plt

category = ["Electronics", "Clothing", "Food", "Books"]
sales = [500, 300, 450, 250]

plt.bar(category, sales,
        color=["blue", "green", "orange", "red"])
plt.title("E-Commerce Sales Analytics")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.show()
```

OUTPUT:



Scenario 7: Satellite Image Vegetation Analysis

R code:

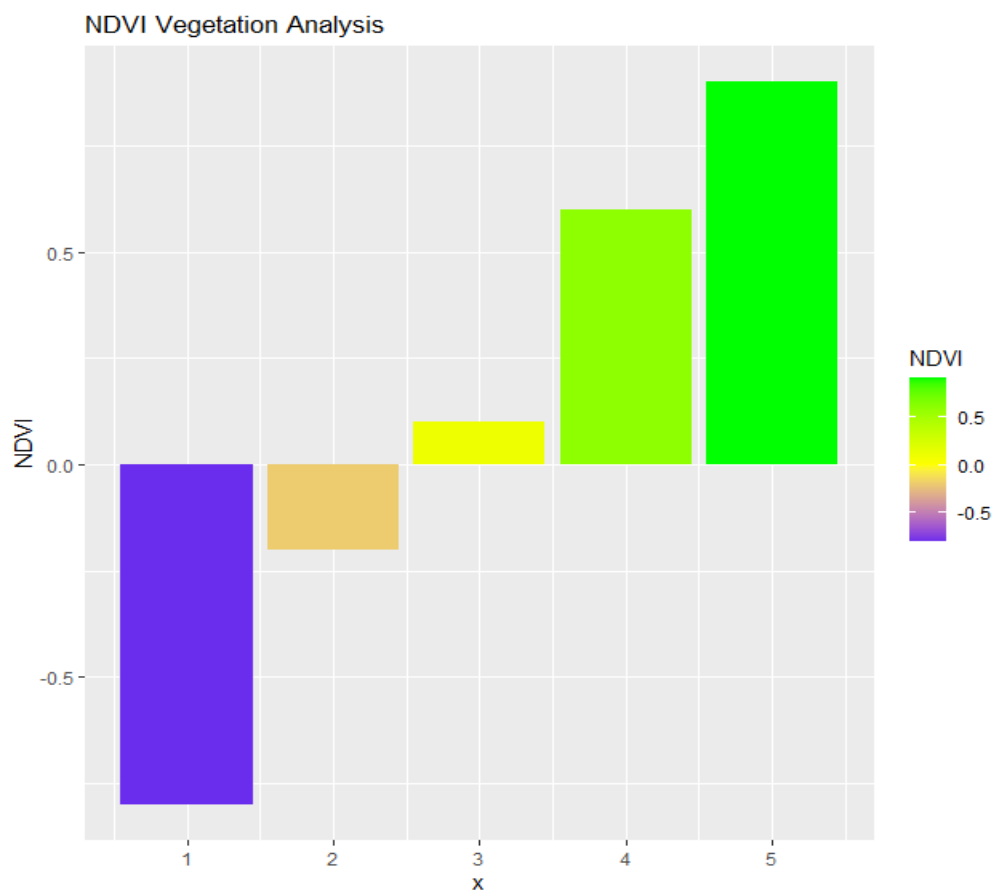
```
library(ggplot2)

x <- 1:5
NDVI <- c(-0.8,-0.2,0.1,0.6,0.9)

data <- data.frame(x, NDVI)

ggplot(data, aes(x, NDVI, fill=NDVI)) +
  geom_col() +
  scale_fill_gradient2(low="blue",
                      mid="yellow",
                      high="green",
                      midpoint=0) +
  ggtitle("NDVI Vegetation Analysis")
```

OUTPUT:



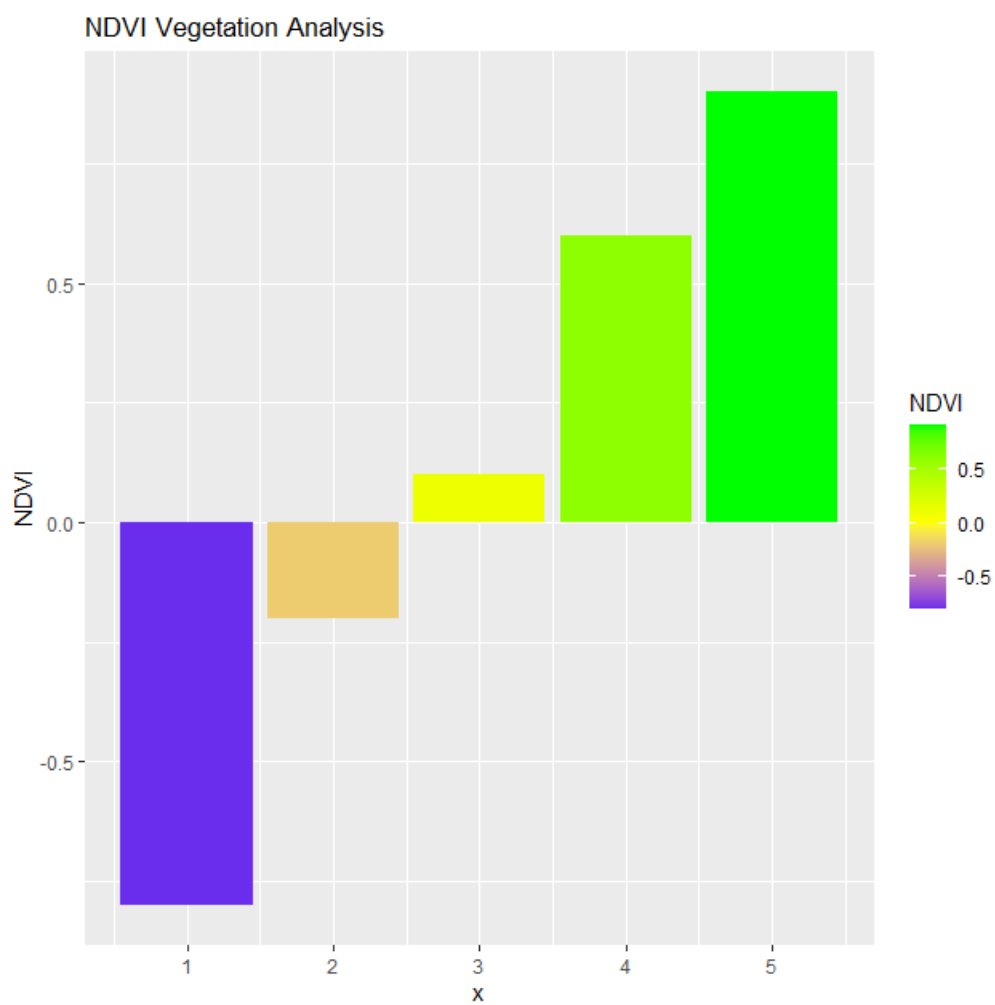
Python code:

```
import matplotlib.pyplot as plt

x = [1,2,3,4,5]
ndvi = [-0.8,-0.2,0.1,0.6,0.9]

plt.bar(x, ndvi,
        color=["blue","cyan","yellow","lightgreen","green"])
plt.title("NDVI Vegetation Analysis")
plt.xlabel("Sample")
plt.ylabel("NDVI")
plt.show()
```

OUTPUT:



Scenario 8: Cryptocurrency Market Volatility

R code:

```
library(ggplot2)

Day <- 1:5
Bitcoin <- c(1000,5000,10000,30000,60000)

data <- data.frame(Day, Bitcoin)

ggplot(data, aes(Day, Bitcoin)) +
  geom_line(color="blue") +
  geom_point(color="red", size=3) +
  scale_y_log10() +
  ggtitle("Cryptocurrency Market Volatility")
```

OUTPUT:



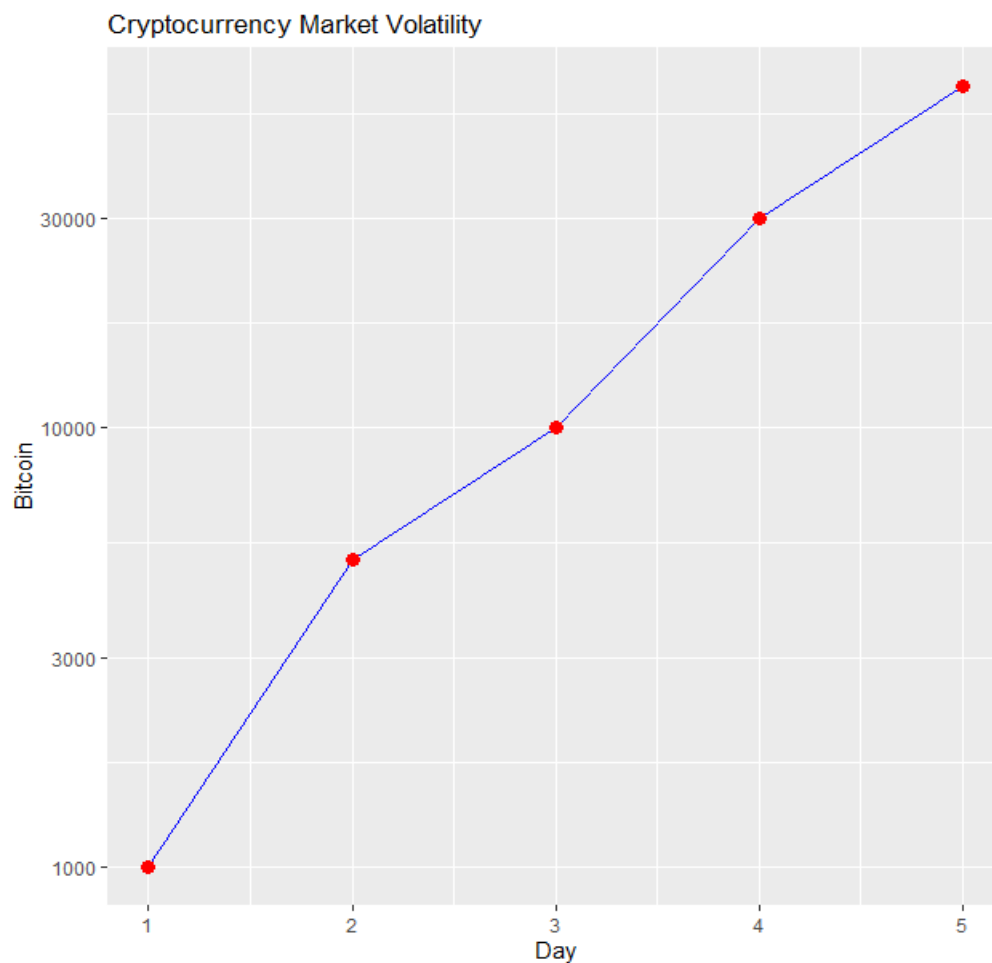
Python code:

```
import matplotlib.pyplot as plt

day = [1,2,3,4,5]
bitcoin = [1000,5000,10000,30000,60000]

plt.plot(day, bitcoin,
         marker="o",
         color="blue")
plt.yscale("log")
plt.title("Cryptocurrency Market Volatility")
plt.xlabel("Day")
plt.ylabel("Bitcoin Price")
plt.grid(True)
plt.show()
```

OUTPUT:



Scenario 9: Marathon Runner Performance Analysis

R code:

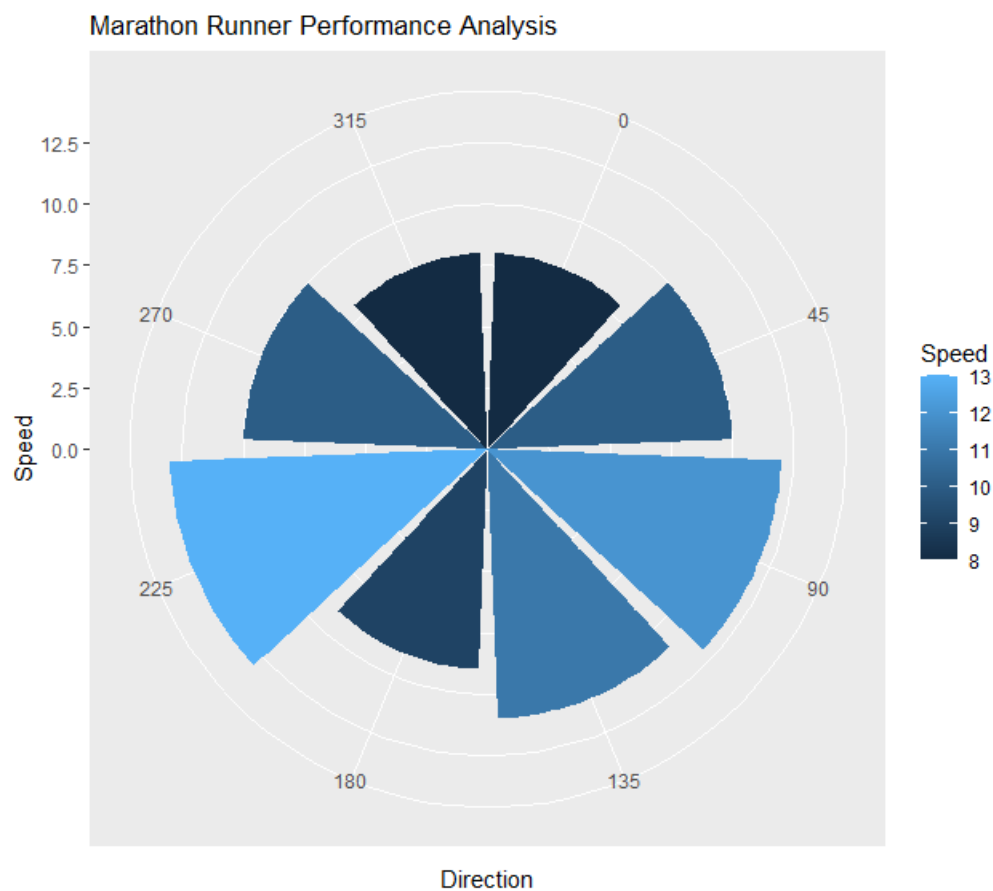
```
library(ggplot2)

Direction <- c(0,45,90,135,180,225,270,315)
Speed <- c(8,10,12,11,9,13,10,8)

data <- data.frame(Direction, Speed)

ggplot(data, aes(x=factor(Direction), y=Speed, fill=Speed)) +
  geom_bar(stat="identity") +
  coord_polar() +
  ggtitle("Marathon Runner Performance Analysis") +
  xlab("Direction") +
  ylab("Speed")
```

OUTPUT:



Python code:

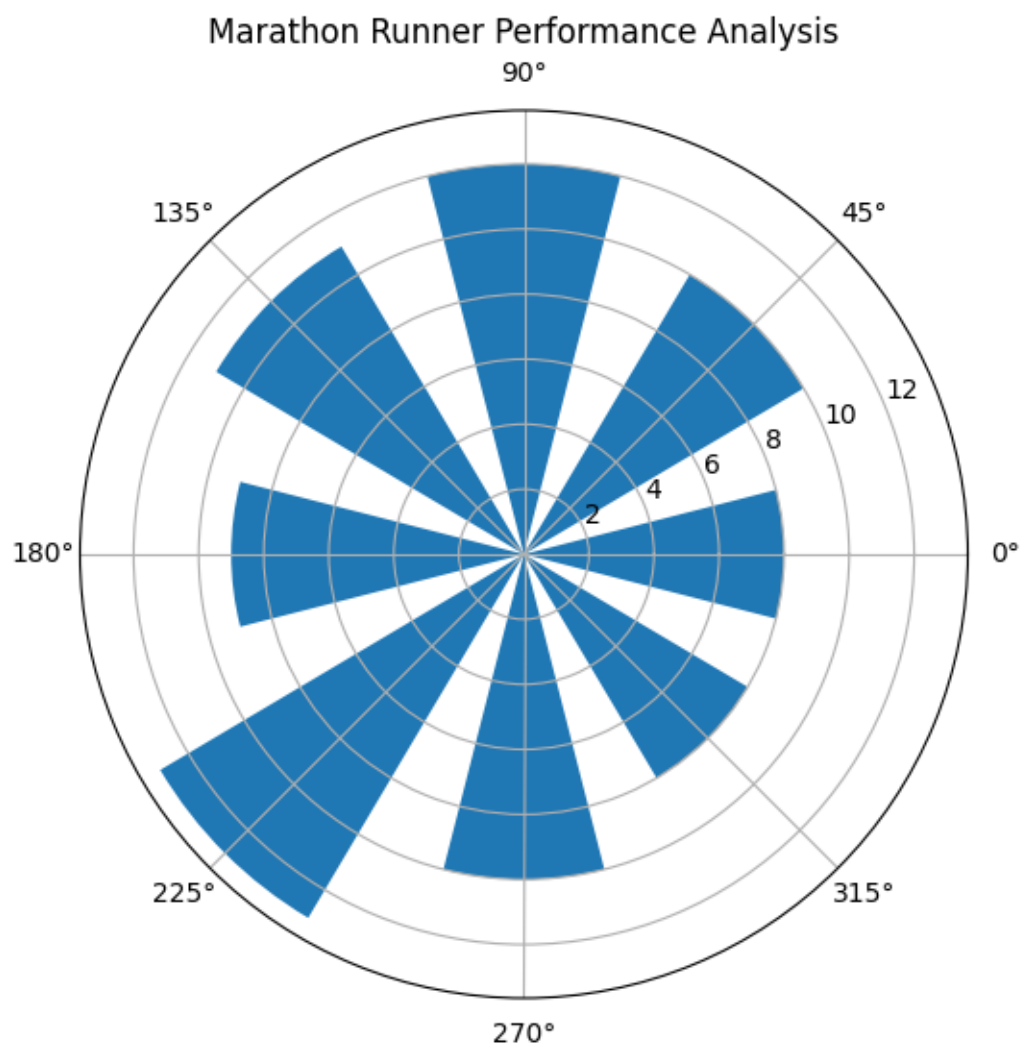
```
import matplotlib.pyplot as plt
import numpy as np

direction = np.deg2rad([0,45,90,135,180,225,270,315])
speed = [8,10,12,11,9,13,10,8]

plt.figure(figsize=(6,6))
ax = plt.subplot(111, polar=True)
ax.bar(direction, speed, width=0.5)

plt.title("Marathon Runner Performance Analysis")
plt.show()
```

OUTPUT:



Scenario 10: COVID-19 Vaccination Campaign Dashboard

R CODE:

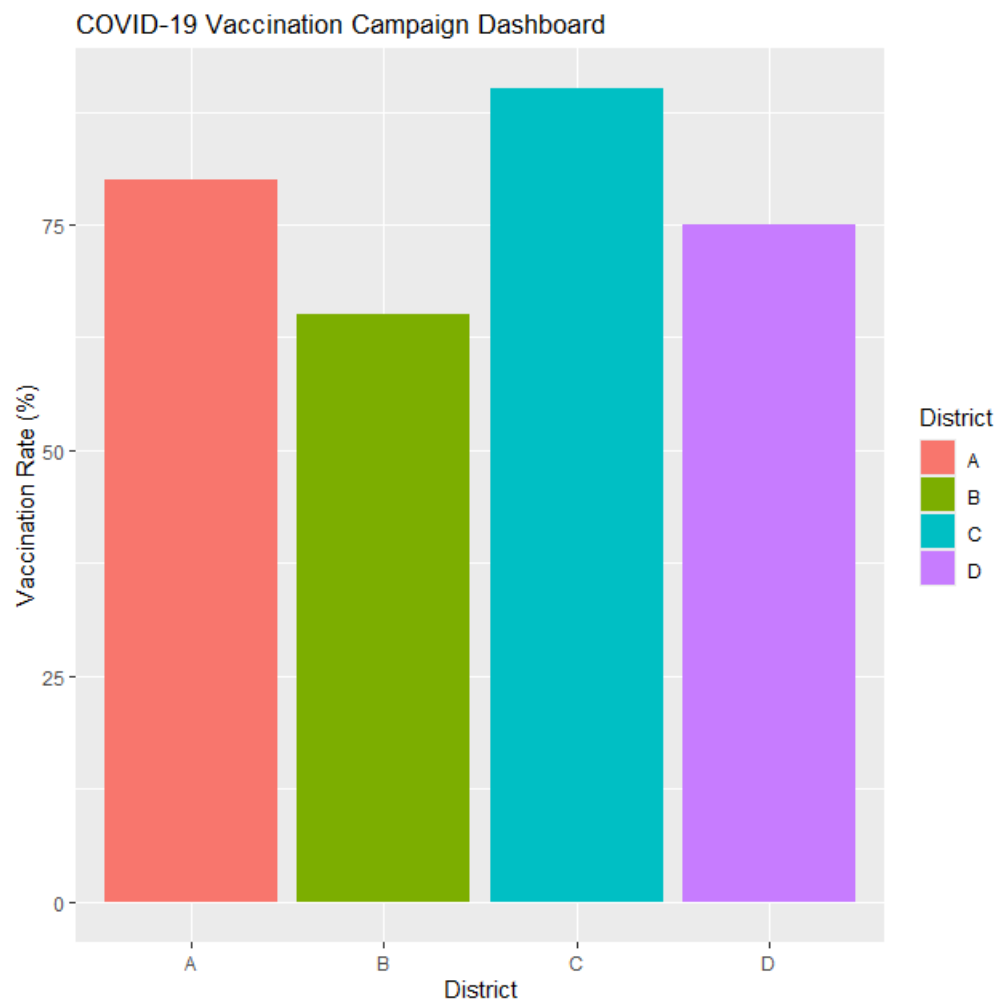
```
library(ggplot2)

District <- c("A","B","C","D")
Vaccination <- c(80,65,90,75)

data <- data.frame(District, Vaccination)

ggplot(data, aes(x=District, y=Vaccination, fill=District)) +
  geom_bar(stat="identity") +
  ggtitle("COVID-19 Vaccination Campaign Dashboard") +
  xlab("District") +
  ylab("Vaccination Rate (%)")
```

OUTPUT:



Python code:

```
import matplotlib.pyplot as plt

district = ["A","B","C","D"]
vaccination = [80,65,90,75]

plt.bar(district, vaccination,
        color=["blue","green","orange","red"])

plt.title("COVID-19 Vaccination Campaign Dashboard")
plt.xlabel("District")
plt.ylabel("Vaccination Rate (%)")
plt.show()
```

OUTPUT:

